

## Lecture 18: Stationary Distribution, cont.

APRIL 13

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## 18.1 Stationary Distribution

### 18.1.1 Generalization to Stationary Distribution for 2-state Markov Chains

Now let's try to use this method to a general 2 state Markov chain, where the two states are "1" and "2". Let  $a$  be the probability of going from 1 to 2, and  $b$  be the probability of going from 2 to 1. Since each row has to sum to 1, this means the general chain looks like

$$\begin{array}{cc} & \begin{array}{cc} \mathbf{1} & \mathbf{2} \end{array} \\ \begin{array}{c} \mathbf{1} \\ \mathbf{2} \end{array} & \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix} \end{array}$$

Now to find the stationary distribution, we form the equations

$$(\pi_1 \quad \pi_2) \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix} = (\pi_1 \quad \pi_2)$$

This means we get the following equations:

$$\begin{aligned} (1-a)\pi_1 + b\pi_2 &= \pi_1 \\ a\pi_1 + (1-b)\pi_2 &= \pi_2 \end{aligned}$$

Since both equations simplify to  $\pi_1 = \frac{b}{a}\pi_2$ , we need to additionally enforce the  $\pi_1 + \pi_2 = 1$  condition to solve the systems. Using substitution, we arrive at

$$\frac{b}{a}\pi_2 + \pi_2 = 1 \Rightarrow \pi_2 = \frac{a}{a+b}$$

, so  $\pi_1 = \frac{b}{a+b}$ . Lastly, we check that this formula is consistent with the answer that we found for the weather chain problem. There  $a = 0.4, b = 0.2$ , so  $\pi_1 = \frac{0.2}{0.4+0.2} = \frac{1}{3}$ , and  $\pi_2 = \frac{0.4}{0.4+0.2} = \frac{2}{3}$ , which is consistent to what we found above.

### 18.1.2 Stationary Distribution for 3- state Markov Chain

Now we turn our attention to 3-state Markov chains, such as the social mobility problem.

$$\begin{aligned}0.7\pi_1 + 0.3\pi_2 + 0.2\pi_3 &= \pi_1 \\0.2\pi_1 + 0.5\pi_2 + 0.4\pi_3 &= \pi_2 \\0.1\pi_1 + 0.2\pi_2 + 0.4\pi_3 &= \pi_3\end{aligned}$$

But observe that if we add the equations together, since the columns come from a stochastic matrix, we get the redundant equation that  $\pi_1 + \pi_2 + \pi_3 = \pi_1 + \pi_2 + \pi_3$ , which will cause the system of equations to be unsolvable. So we need to change this to  $\pi_1 + \pi_2 + \pi_3 = 1$ , since it is a probability distribution, and it must sum to 1. This means that we get the following system of equations.

$$\begin{aligned}-0.3\pi_1 + 0.3\pi_2 + 0.2\pi_3 &= 0 \\0.2\pi_1 - 0.5\pi_2 + 0.4\pi_3 &= 0 \\\pi_1 + \pi_2 + \pi_3 &= 1\end{aligned}$$

We note that the third equation implies  $\pi_3 = 1 - \pi_1 - \pi_2$  and substituting this in the first two gives

$$\begin{aligned}0.2 &= -0.5\pi_1 - 0.1\pi_2 \\0.4 &= 0.2\pi_1 + 0.9\pi_2\end{aligned}$$

Multiplying the first equation by 0.9 and adding 0.1 times the second gives

$$2.2 = (0.45 + 0.02)\pi_1 \implies \pi_1 = \frac{22}{47}.$$

Multiplying the first equation by 0.2 and adding -0.5 times the second gives

$$-0.16 = (-0.02 - 0.45)\pi_2 \implies \pi_2 = \frac{16}{47}.$$

Since the three probabilities add up to 1,  $\pi_3 = \frac{9}{47}$ .

This might have been hard with all of the computation, so let's make a computer do the hard work. If we take the above system of equations, we can rewrite it as a matrix equation:

$$\begin{cases} -0.3\pi_1 + 0.3\pi_2 + 0.2\pi_3 = 0 \\ 0.2\pi_1 - 0.5\pi_2 + 0.4\pi_3 = 0 \\ \pi_1 + \pi_2 + \pi_3 = 1 \end{cases} \implies \begin{pmatrix} \pi_1 & \pi_2 & \pi_3 \end{pmatrix} \begin{pmatrix} -0.3 & 0.2 & 1 \\ 0.3 & -0.5 & 1 \\ 0.2 & 0.4 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$$

This means to solve this problem, we need to solve this equation: Let

$$\mathbf{A} = \begin{pmatrix} -0.3 & 0.2 & 1 \\ 0.3 & -0.5 & 1 \\ 0.2 & 0.4 & 1 \end{pmatrix}.$$

Then we have  $\pi\mathbf{A} = (0 \ 0 \ 1)$ , so we need to multiply both sides by  $\mathbf{A}^{-1}$  on the right to get

$$\begin{pmatrix} \pi_1 & \pi_2 & \pi_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \mathbf{A}^{-1}.$$

Using MATLAB, we can easily compute this to get

$$\pi = (0.468085 \quad 0.340425 \quad 0.191489) = \left(\frac{22}{47} \quad \frac{16}{47} \quad \frac{9}{47}\right).$$

**Example 18.1.** Suppose there are three types of laundry detergent, 1, 2, and 3, and let  $X_n$  be the brand chosen on the  $n^{\text{th}}$  purchase. Customers who try these brands are satisfied and choose the same thing again with probabilities 0.8, 0.6, and 0.4 respectively. When they change they pick one of the other two brands at random. The transition probability is

|   | 1   | 2   | 3   |
|---|-----|-----|-----|
| 1 | 0.8 | 0.1 | 0.1 |
| 2 | 0.2 | 0.6 | 0.2 |
| 3 | 0.3 | 0.3 | 0.4 |

Using the first two equations and the fact that the sum of the  $\pi_i$ 's is 1:

$$0.8\pi_1 + 0.2\pi_2 + 0.3\pi_3 = \pi_1$$

$$0.1\pi_1 + 0.6\pi_2 + 0.3\pi_3 = \pi_2$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

Subtracting  $\pi_1$  from both sides of the first equation and  $\pi_2$  from both sides of the second, this translates into  $\pi\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$ , with

$$\begin{pmatrix} -0.2 & 0.1 & 1 \\ 0.2 & -0.4 & 1 \\ 0.3 & 0.3 & 1 \end{pmatrix}$$

Note that here and in the previous example the first two columns of  $\mathbf{A}$  consist of the first two columns of the transition probability with 1 subtracted from the diagonal entries, and the final column is all 1's. Computing the inverse and reading the last row gives

$$\pi = (0.545454 \quad 0.272727 \quad 0.181818) = \left(\frac{6}{11} \quad \frac{3}{11} \quad \frac{2}{11}\right).$$

To check this we note that

$$\begin{pmatrix} \frac{6}{11} & \frac{3}{11} & \frac{2}{11} \end{pmatrix} \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.3 & 0.3 & 0.4 \end{pmatrix} = \begin{pmatrix} \frac{4.8+0.6+0.6}{11} & \frac{0.6+1.8+0.6}{11} & \frac{0.6+0.6+0.8}{11} \end{pmatrix}$$

## 18.2 Limit Behavior

**Remark 18.2.** If  $y$  is transient, then as you take more and more steps, the probability of returning to that state gets smaller and smaller until it becomes zero. Mathematically, we have  $p^n(x, y) \rightarrow 0$ .

This means that we can restrict our attention to recurrent states and in view of the decomposition theorem, Theorem 1.8, to chains that consist of a single irreducible class of recurrent states. Our first example shows one problem that can prevent the convergence of  $p^n(x, y)$ .

**Example 18.3** (Ehrenfest chain). This chain originated in physics as a model for two cubical volumes of air connected by a small hole. In the mathematical version, we have two “urns,” i.e., two of the exalted trash cans of probability theory, in which there are a total of  $N$  balls. We pick one of the  $N$  balls at random and move it to the other urn. Let  $X_n$  be the number of balls in the “left” urn after the  $n^{\text{th}}$  draw. It should be clear that  $X_n$  has the Markov property; i.e., if we want to guess the state at time  $n + 1$ , then the current number of balls in the left urn  $X_n$ , is the only relevant information from the observed sequence of states  $X_n, X_{n-1}, \dots, X_1, X_0$ . To check this we note that:

$$\mathbb{P}(X_{n+1} = i + 1 | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = \frac{N - i}{N}$$

since to increase the number we have to pick one of the  $N - i$  balls in the other urn. The number can also decrease by 1 with probability  $i/N$ . In symbols, we have computed that the transition probability is given by

$$p(i, i + 1) = \frac{N - i}{N}, \quad p(i, i - 1) = \frac{i}{N} \quad \text{for } 0 \leq i \leq N.$$

with  $p(i, j) = 0$  otherwise. When  $N = 3$ , we get

|   | 0   | 1   | 2   | 3   |
|---|-----|-----|-----|-----|
| 0 | 0   | 3/3 | 0   | 0   |
| 1 | 1/3 | 0   | 2/3 | 0   |
| 2 | 0   | 2/3 | 0   | 1/3 |
| 3 | 0   | 0   | 3/3 | 0   |

In the second power of  $p$  the zero pattern is shifted:

|   | 0   | 1   | 2   | 3   |
|---|-----|-----|-----|-----|
| 0 | 1/3 | 0   | 2/3 | 0   |
| 1 | 0   | 7/9 | 0   | 2/9 |
| 2 | 2/9 | 0   | 7/9 | 0   |
| 3 | 0   | 2/3 | 0   | 1/3 |

To see that the zeros will persist, note that if we have an odd number of balls in the left urn, then no matter whether we add or subtract one the result will be an even number. Likewise, if the number is even, then it will be odd on the next one step. This alternation between even and odd means that it is impossible to be back where we started after an odd number of steps. In symbols, if  $n$  is odd then  $p^n(x, x) = 0$  for all  $x$ .

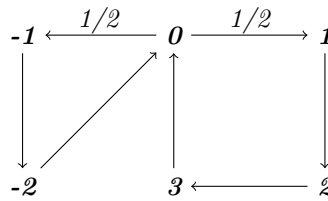
**Definition 18.4.** The period of a state is the largest number that will divide all the  $n \geq 1$  for which  $p^n(x, x) > 0$ . That is, it is the greatest common divisor of  $I_x = \{n \geq 1 : p^n(x, x) > 0\}$ .

**Remark 18.5.** To check that this definition works correctly, we note that in Example 1.21,  $\{n \geq 1 : p^n(x, x) > 0\} = \{2, 4, \dots\}$ , so the greatest common divisor is 2.

**Example 18.6.** Triangle and Square

|    | -2 | -1  | 0 | 1   | 2 | 3 |
|----|----|-----|---|-----|---|---|
| -2 | 0  | 0   | 1 | 0   | 0 | 0 |
| -1 | 1  | 0   | 0 | 0   | 0 | 0 |
| 0  | 0  | 0.5 | 0 | 0.5 | 0 | 0 |
| 1  | 0  | 0   | 0 | 0   | 0 | 1 |
| 2  | 0  | 0   | 0 | 0   | 0 | 0 |
| 3  | 0  | 0   | 1 | 0   | 0 | 0 |

In words, from 0 we are equally likely to go to 1 or -1. From -1 we go with probability one to -2 and then back to 0, from 1 we go to 2 then to 3 and back to 0. The name refers to the fact that  $0 \rightarrow -1 \rightarrow -2 \rightarrow 0$  is a triangle and  $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$  is a square



Clearly,  $p^3(0,0) > 0$  and  $p^4(0,0) > 0$  so  $3, 4 \in I_0$ .

We need to think to realized that

$$I_0 = \{3, 4, 5, 6, 7, 8, 9, 10, 11, \dots\}$$

While periodicity is a theoretical possibility, it rarely manifests itself in applications, except occasionally as an odd-even parity problem, e.g., the Ehrenfest chain. In most cases we will find (or design) our chain to be aperiodic, i.e., all states have period 1. To be able to verify this property for examples, we need to discuss some theory.

**Lemma 18.7.** *If  $p(x,x) > 0$ , then  $x$  has period 1.*

**Lemma 18.8.** *If  $\rho_{xy} > 0$  and  $\rho_{yx} > 0$  then  $x$  and  $y$  have the same period.*

**Remark 18.9.** *The short answer is that if the two states have different periods, then by going from  $x$  to  $y$ , from  $y$  to  $y$  in the various possible ways, and then from  $y$  to  $x$ , we will get a contradiction.*

**Theorem 18.10. Convergence Theorem** *Let  $p$  be irreducible, aperiodic, and there is a stationary distribution  $\pi$ . Then as  $n \rightarrow \infty$ ,  $p^n(x,y) \rightarrow \pi(y)$ .*